

Samuele Tondelli

MSc Student | ML Systems & GPU Performance

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SUMMARY

Final-year master's student in Computer Science and Engineering focused on ML systems and GPU performance. Optimizing graph transformer inference for protein docking with CUDA, Triton, and TensorRT. Seeking internship and research opportunities from November 2026 to July 2027.

EDUCATION

Politecnico di Milano 2024 – July 2027 (expected)
Master in Computer Science and Engineering
Currently enrolled.

University of Modena and Reggio Emilia 2021 – 2024
Bachelor in Computer Engineering
Thesis: "Optimization and synthesis on FPGA of the logarithm of softmax function for neuromorphic accelerators". 110 cum laude.

PROJECTS

GPU ACCELERATED GRAPH TRANSFORMER INFERENCE 2026
Master's thesis | Politecnico di Milano
Optimizing GPU inference of a graph transformer for protein docking, in active development.

- Implemented LibTorch C++ and TensorRT backends running on an NVIDIA A100.
- Profiled with Nsight Systems; removed host-device synchronization to cut forward-pass latency from 63 ms to 12 ms.
- Fused full graph attention in Triton using sparse CSC indices, cutting latency from 190 us to 46 us vs. PyTorch.

Technologies: GPU, CUDA, Triton, PyTorch, TensorRT, Graph Transformer, Nsight

GRADINO - OPTIMIZED C AUTOGRAD ENGINE 2026
Autograd engine written in C optimized for CPU training with SIMD acceleration and custom allocators.
github.com/SamueleTondelli/gradino
Technologies: C, Deep learning, SIMD, AVX-512, Arena allocators

EXPERIENCE

Student Researcher (Genomic ML) Sep 2025 – Feb 2026
NECSTLab, Politecnico di Milano Italy

- Conducted a comparative study of ML approaches for single nucleotide variant classification.
- Benchmarked 4 models (MLP, XGBoost, Logistic Reg, Random Forest) across 144 genomic datasets.
- Reduced training time (24 s to 2.6 s) and peak RAM (2.28 GB to 0.30 GB), with minimal test-performance impact.
- Measured 90% average pairwise agreement on VUS pathogenicity across 72 models trained without VUS.

Technologies: ML, Genomics, Python, Hydra, Scikit-learn

Student Researcher (FPGA Deep Space) Mar 2025 – Sep 2025
NECSTLab, Politecnico di Milano Italy

- Accelerated ECOS library for autonomous navigation of deep space satellites.
- Developed accelerator for sparse matrix computations (SpMV and LDLt) using Vitis HLS and Vivado.
- Deployed IP on Kria KD240 and developed a C Linux driver, achieving 2x speedup over software.
- Competed in the 2025 edition of the AMD Open Hardware competition.

Technologies: FPGA, Vitis HLS, Vitis, Vivado, Xilinx, Embedded systems, C

Software Engineer Intern (Formula SAE) Mar 2023 – May 2024
More Modena Racing, UniMoRe Italy

- Co-led a team of 4 developing software for combustion and hybrid Formula SAE cars.
- Developed live, wireless telemetry (1 Mbps over 500m) using Raspberry Pi 5 and Grafana.
- Developed car dashboard software on STM32 controller reading CAN bus data.

Technologies: Automotive, C, Raspberry Pi, Live telemetry, Grafana, STM32, CAN

PUBLICATIONS

A Comparative Analysis of Machine Learning Methods for Genomic Variants Classification.
Samuele Tondelli, Riccardo Masetti, Leonardo De Grandis, Guido Walter Di Donato, Marco Domenico Santambrogio.
IEEE EMBC 2026. Accepted paper, Paper No. 3391.

SKILLS

GPU & performance: CUDA, SIMD, Triton, AVX-512, TensorRT, Nsight Systems/Compute
Programming: C, C++, Python, Rust, Assembly, Go, Java
ML frameworks: PyTorch, Scikit-learn, Hydra, TensorFlow
Dev tools: Linux, Git, Make, CMake, GCC, GDB, Valgrind, Vivado, ModelSim
Hardware & embedded: Vitis HLS, STM32, Verilog, Xilinx Zynq FPGAs, Raspberry Pi
Languages: English (C1, certified); Italian